

Future of Physical Infrastructures

1. Introduction– The Problem we all Face

You're in the middle of an intense Valorant match, and boom — your GPU dies. Game over. You're staring at a black screen, stuck with a dead piece of hardware. You search for a cloud gaming service to keep going, but they're either too expensive, laggy, or not available in your region.

Now imagine a world where someone nearby is sharing their unused GPU power — maybe even your old one — and you can tap into it instantly. No middlemen, no fat subscription fees, just peer-to-peer access to real resources. That's the power of DePIN.

2. What is DePIN? (Decentralized Physical Infrastructure Networks)

DePIN stands for **Decentralized Physical Infrastructure Networks**. In simple terms, it's a way for **real-world infrastructure** — like storage devices, wireless hotspots, GPUs, or even dashcams — to be shared by everyday people rather than big corporations.

Anyone can become a provider by contributing their unused resources. In return, they get paid. Users get access to cheaper, faster, and decentralized services. All of this runs through blockchain smart contracts that automate payments and verify usage — no need for trust or a middleman.

Think of it like Uber or Airbnb, but instead of rides or homes, you're renting **computing power, internet coverage, or even maps** — and it's fully decentralized.

3. Why DePIN on Solana? Why Not Others?

For DePIN to work at scale, you need speed, low fees, and the ability to handle a massive number of users and devices — all interacting in real time. That's where **Solana** shines.

Here's why Solana is the perfect match for DePIN:

- **Ultra-low fees**

DePIN often requires **microtransactions** — tiny payments happening frequently (like paying for GPU usage per second or per MB of data). On Solana, fees are fractions of a cent, while networks like Ethereum can charge dollars per transaction.

- **High throughput**

Solana can handle **thousands of transactions per second**, making it ideal for live infrastructure usage like cloud GPUs, hotspots, or real-time map data. Ethereum or even Layer-2s often face congestion or latency issues.

- **Fast finality**

Solana finalizes transactions in seconds, so if you're streaming from someone's GPU or using their hotspot, you don't experience lag due to blockchain delays. You can't say the same about most other chains.

- **Scalable without rollups**

While Ethereum relies on rollups and external layers for scalability (which adds complexity and trust assumptions), Solana scales at the base layer, making it **simpler, faster, and more secure**.

4. Why Solana DePin is better than the Traditional Web2:

Traditional Web2 Infra: Centralized & Controlled	Solana DePIN: Power to the People
 Big Tech Monopolies: A few giants own all the servers, data, and decisions.	 Truly Decentralized: Anyone can contribute hardware (like GPUs, storage, sensors) and earn.
 You Pay, They Profit: You buy storage or compute power—they earn recurring revenue, you own nothing .	 Earn-as-You-Contribute: Your idle resources become a passive revenue-generating assets.
 Walled Gardens: Closed ecosystems, limited control, and zero transparency .	 Open & Permissionless: No gatekeepers. Anyone, anywhere can use or provide infra.
 Limited Accessibility : Only the handful of people who are there can use it- not for everyone in the world.	 Scalable & Sustainable: Grows as more people join— not limited by centralized data centers .

5. Some Solana-based DePIN Projects

i) Shaga – Rent Out Your Gaming PC

Got a gaming rig sitting idle?

Shaga lets you rent out your unused GPU power to people who need it — for AI training, rendering, or gaming — all through a secure, decentralized network



“With Shaga, my idle PC paid for my internet bill. It’s like turning my setup into a passive income machine.”

— A real DePIN provide



ii) Helium – The Decentralized Wireless Network

Helium is the “OG” DePIN project. People run **hotspots** that provide wireless coverage (LoRaWAN

and 5G) in their areas — and they get paid for it. Businesses and IoT devices connect to this network instead of traditional telecom providers.



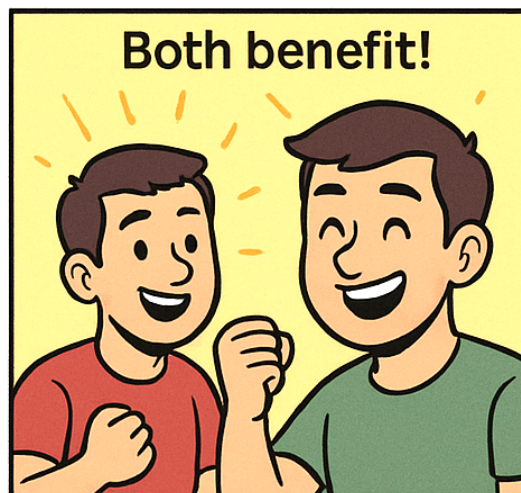
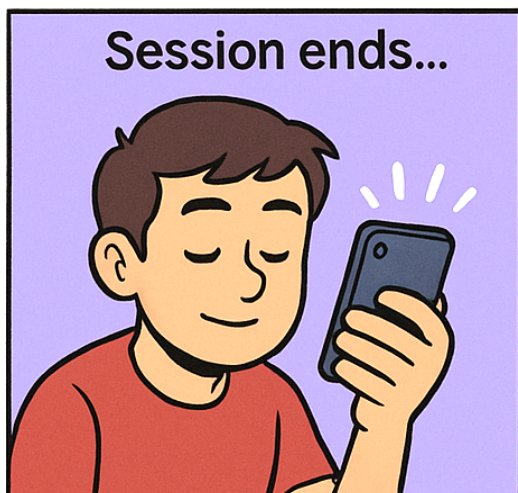
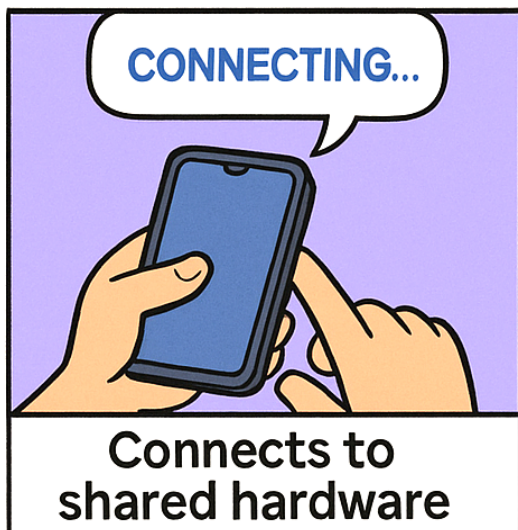
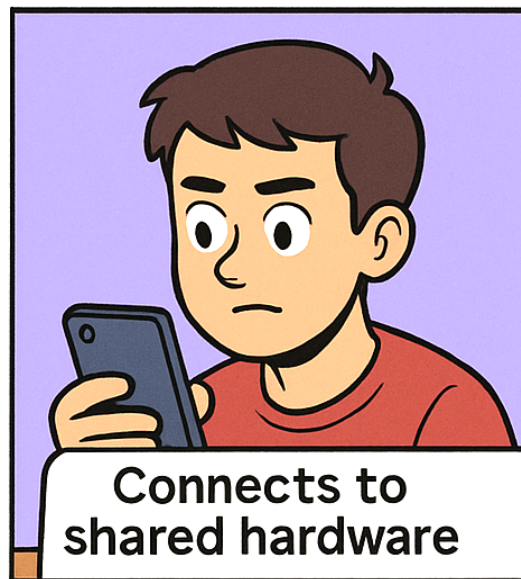
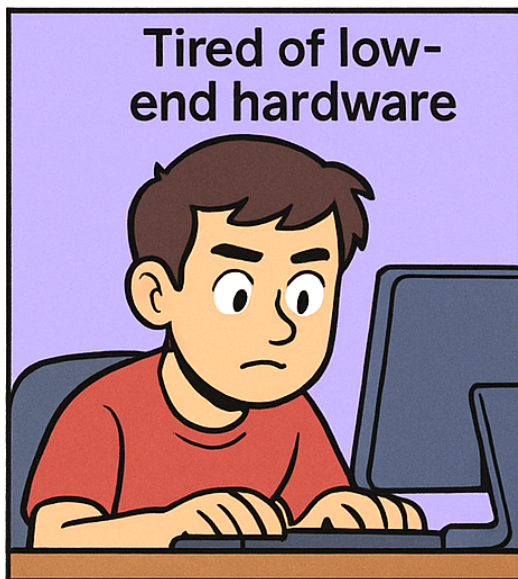
iii) Cudis- AI Powered Ring

Tracks your movement, sleep, stress, and energy levels in real-time with cutting-edge biometrics.

Designed like a luxury ring, but packed with sensors. Users get rewarded for healthy habits.

“The first AI smart ring that rewards your wellness journey.” 💍

6) How DePIN Works (For Users & Providers)



7). Security Concerns & Trust Issues:

? Malicious Activity: What if a provider installs spyware in their hardware? Or a user exploits the device for illegal activities?

✓ Encryption + Smart Contracts: Prevent unauthorized access and set rules for interaction.

? Network Gaps: What if there are no providers around you? Can the system survive in low-population areas?

✓ Fallbacks: Users can choose a default to cloud/centralized options temporarily if local network fails. Or can switch to some other nearby networks with permission.

? Trust Issues: Devices in the network are owned by random individuals—not big trusted companies. That’s both a strength *and* a risk.

✓ Decentralized Auditing: Community watchdogs and automated checks spot unusual behavior.

Final Words

We’re on the verge of entering a world where you don’t need some big tech company dictating what products you can use, what services you must follow, or how to spend your own money.

Here, *you* are the defining factor.

You decide. You build. You distribute.

And thousands like you will make this movement broader, stronger, and truly unstoppable.

This is not just a new kind of infrastructure — it's a new kind of internet.

Built by the people. Owned by the people.

Powered by Solana.